DevOps Fundamentals Mini Project

Cloud-Based Linux Server Deployment with GitHub Version Control and Automation

Name: Kunal Dait

Batch Code: A530

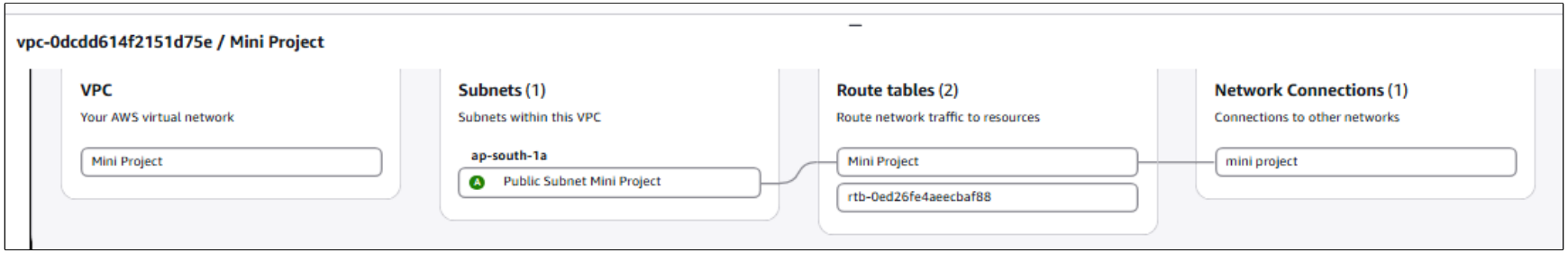
Project Overview

- ABC Technologies needs its internal project website hosted on an AWS cloud server
- Role: Junior DevOps Engineer
- Built the cloud network, configured the Linux server and deployed the website
- Automated routine tasks with Bash scripts and maintained the project on GitHub

Tasks completed

1. AWS infrastructure setup
2. Ubuntu server configuration
3. Website deployment
4. Git and GitHub
5. Bash automation scripts
6. Log monitoring
7. Documentation

Task 1: AWS Infrastructure Setup

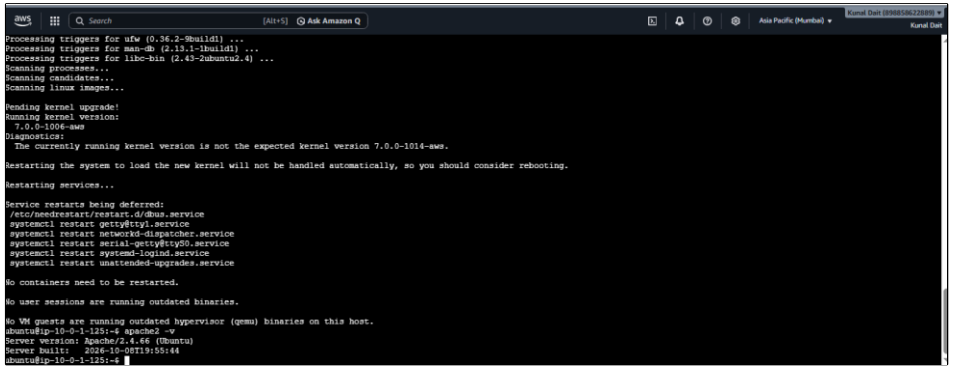


Component	Configuration
VPC	Mini Project
Public subnet	Public Subnet Mini Project (ap-south-1a)
Internet gateway	mini project, attached to the VPC
Route table	Mini Project, routes traffic to the internet gateway
EC2 instance	Ubuntu server in the public subnet, connected over SSH

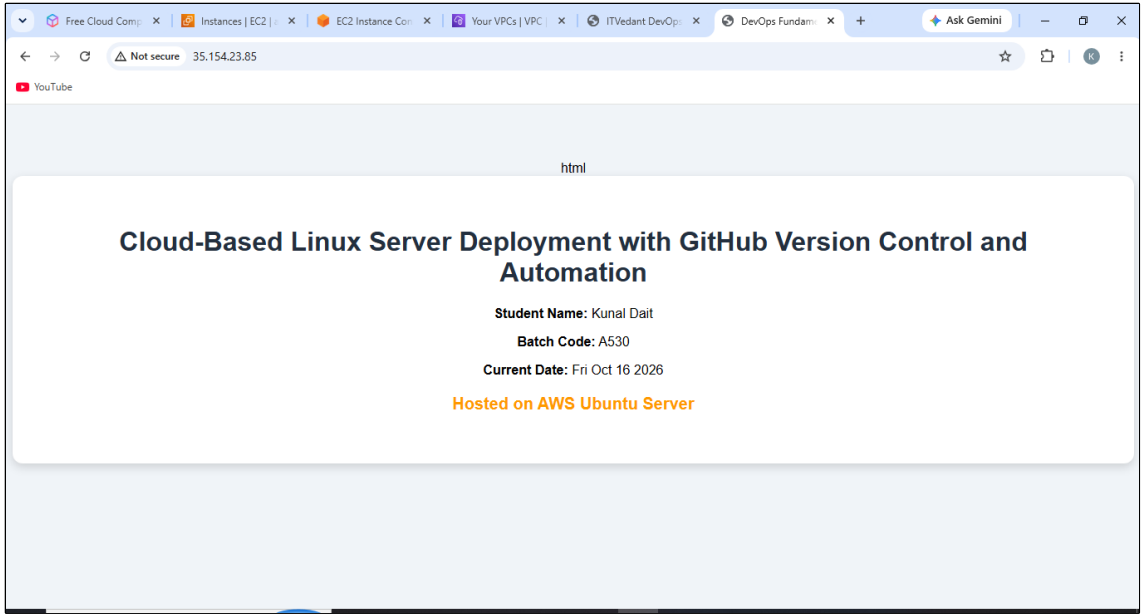
Task 2: Ubuntu Server Configuration

- Updated and upgraded packages with apt
- Installed Git, Curl, Vim, Nginx and Apache
- Verified versions: Git 2.53.0, Curl 8.18.0, Nginx 1.28.3, Apache 2.4.66
- Nginx serves the website on port 80



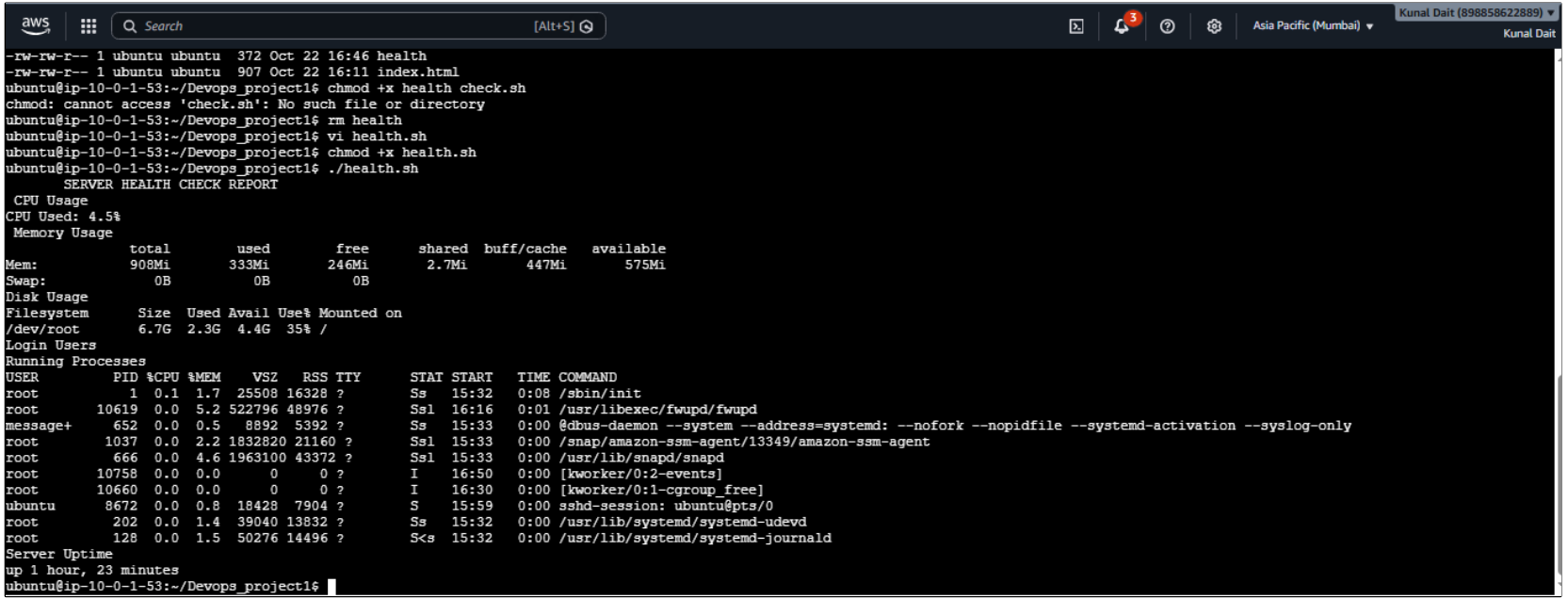


Task 3: Website Deployment



- Opened at http://35.154.23.85
- Shows project title, student name, batch code and date
- Message: Hosted on AWS Ubuntu Server

Task 5: Server Health Check Script



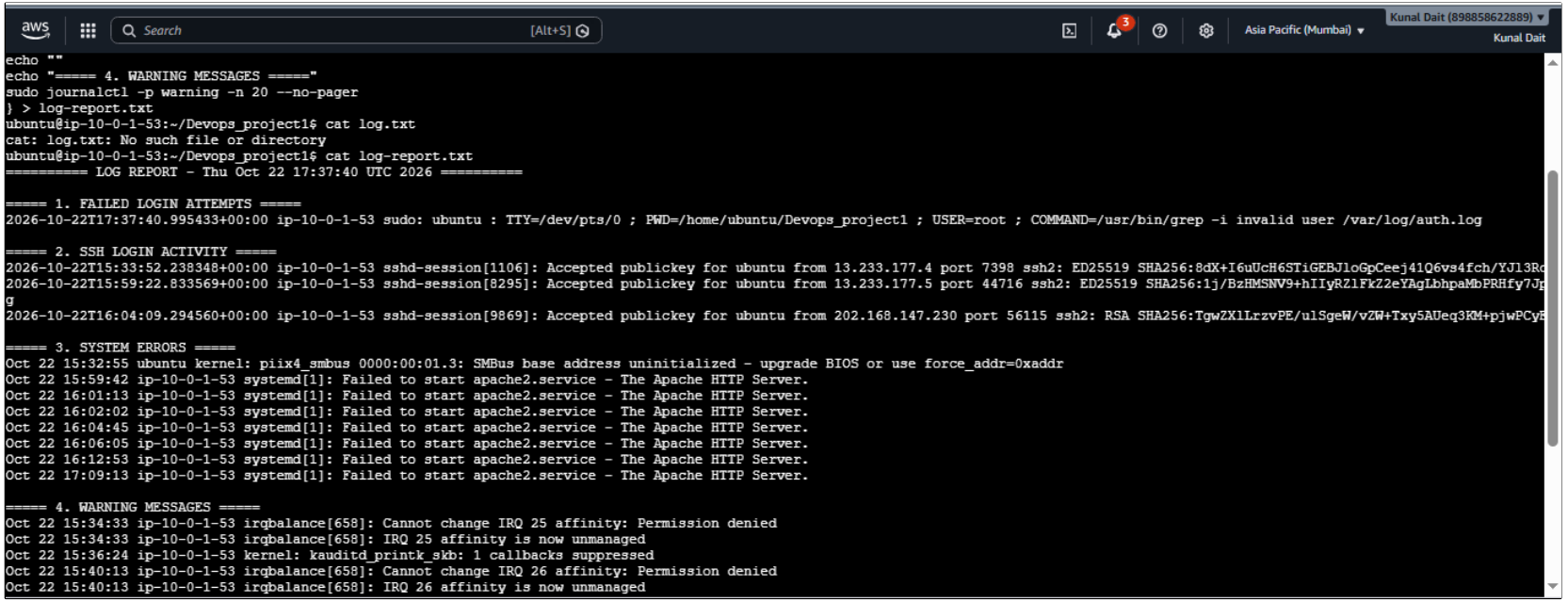
health.sh reports CPU usage (4.5%), memory (333Mi of 908Mi used), disk (35% used), running processes and server uptime.

Task 5: Service Monitor and Backup Scripts



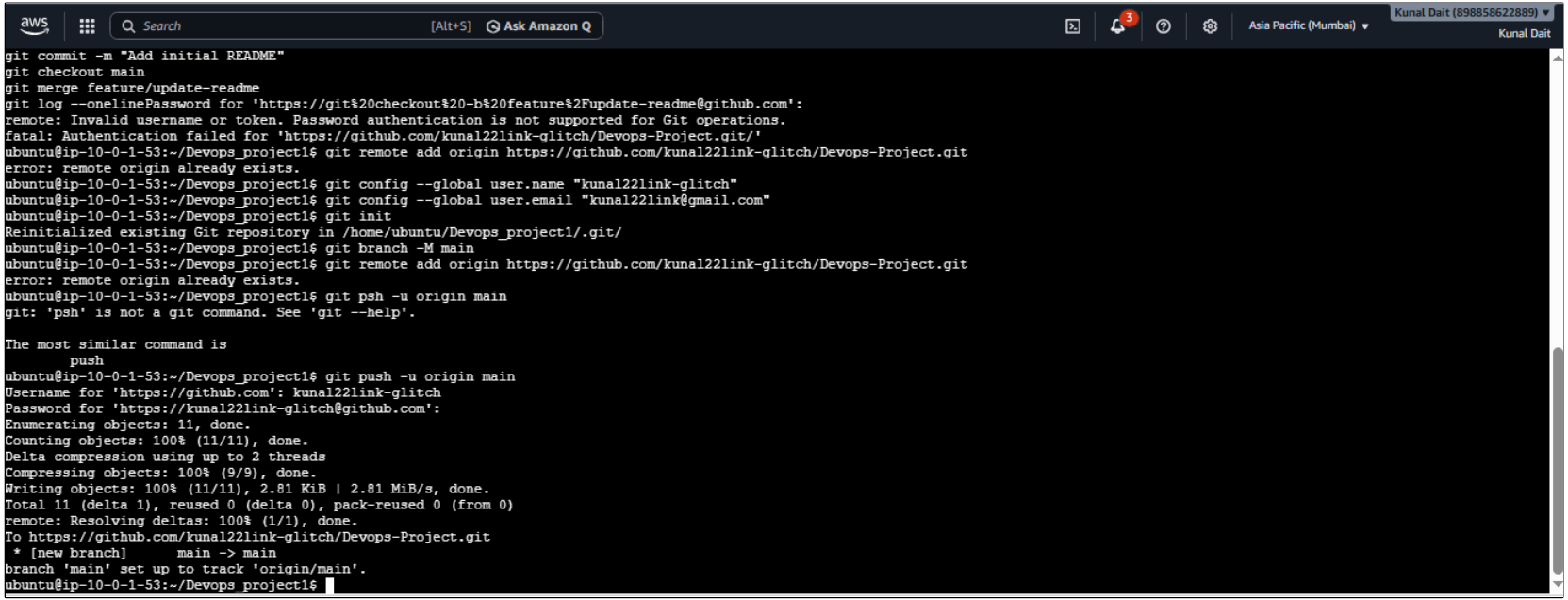
monitor.sh found nginx running, so no action was needed. backup.sh saved a timestamped archive in /home/ubuntu/Backup.

Task 6: Log Monitoring Report



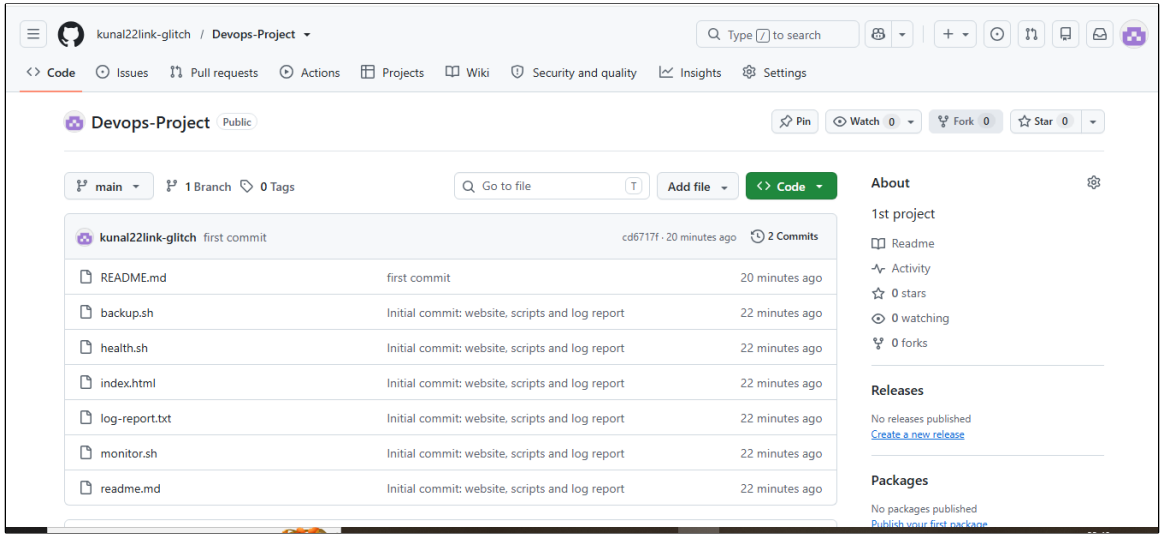
log-report.txt covers failed logins, SSH activity, system errors and warning messages.

Task 4: Git Operations



Commands used: git init, git branch -M main, git merge feature/update-readme, git remote add origin, git push -u origin main.

Task 4: GitHub Repository



- Repository: Devops-Project (Public)
- Owner: kunal22link-glitch
- Contains website, scripts and log report
- Changes pushed to the main branch

Folder Structure

```
Devops-Project/
|-- README.md
|-- index.html
|-- health.sh
|-- monitor.sh
|-- backup.sh
|-- log-report.txt
|-- screenshots/
|   |-- aws-setup.png
|   |-- website.png
|   |-- github.png
|-- documentation/
    |-- Project_Report.pdf
```

Learning Outcomes

- Built a custom AWS network: VPC, subnet, internet gateway and route table
- Launched and accessed an Ubuntu EC2 server over SSH
- Installed packages and managed services on Linux
- Deployed a website and verified it from the public IP
- Wrote Bash scripts for health checks, service monitoring and backups
- Analyzed Linux logs to find errors, warnings and login activity
- Used Git branching, merging and GitHub for version control

Thank You